

DIGITAL IMAGE PROCESSING

LECTURE # 6

IMAGE ENHANCEMENT IN SPATIAL DOMAIN-II

Contact Information

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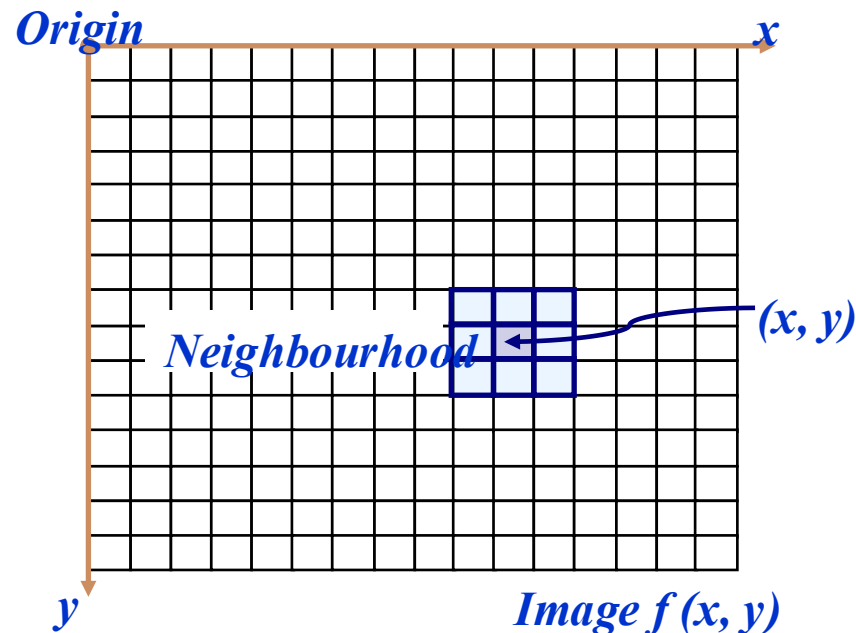
Topics to Cover

- Local/ Neighborhood Operations on images
- Local Enhancement through Spatial Filtering
- Smoothing Spatial Filtering
 - ❑ Linear Filters
 - ❑ Non-Linear Filters
- Linear Smoothing Spatial Filtering
 - ❑ Average Filters
 - ❑ Weighted Average Filters
- Non-Linear Smoothing Spatial Filtering
 - ❑ Median Filters
 - ❑ Minimum Filters
 - ❑ Maximum Filters
- Border Pixel Treatment
- Image Histogram
- Histogram Equalization

Neighborhood Operations on Images

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- Neighbourhood operations simply operate on a larger neighbourhood of pixels than point operations
- Neighbourhoods are mostly selected as square regions around a central pixel
- Any size and any shape filter are possible



Local Operations through Spatial Filtering

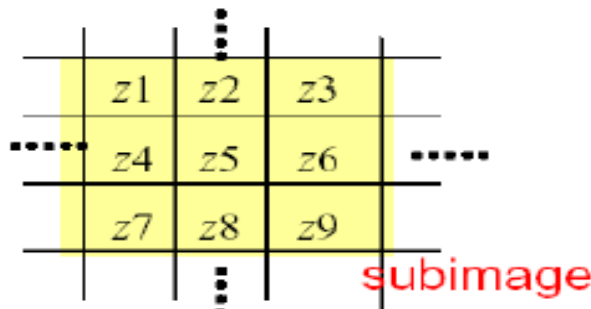
5

- ❑ The output intensity value at (x,y) depends not only on the input intensity value at (x,y) but also on the specified number of neighboring intensity values around (x,y)
- ❑ **Spatial masks** (also called window, filter, kernel, template) are used and **convolved over the entire image for local enhancement** (spatial filtering)
- ❑ The size of the mask determines the number of neighboring pixels which influence the output value at (x,y)
- ❑ The values (coefficients) of the mask determine the nature and properties of enhancing technique

Basics of Spatial Filtering

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- Given the 3×3 mask with coefficients: $w_1, w_2, \dots, w_9$
- The mask covers the pixels with gray levels: $z_1, z_2, \dots, z_9$



w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

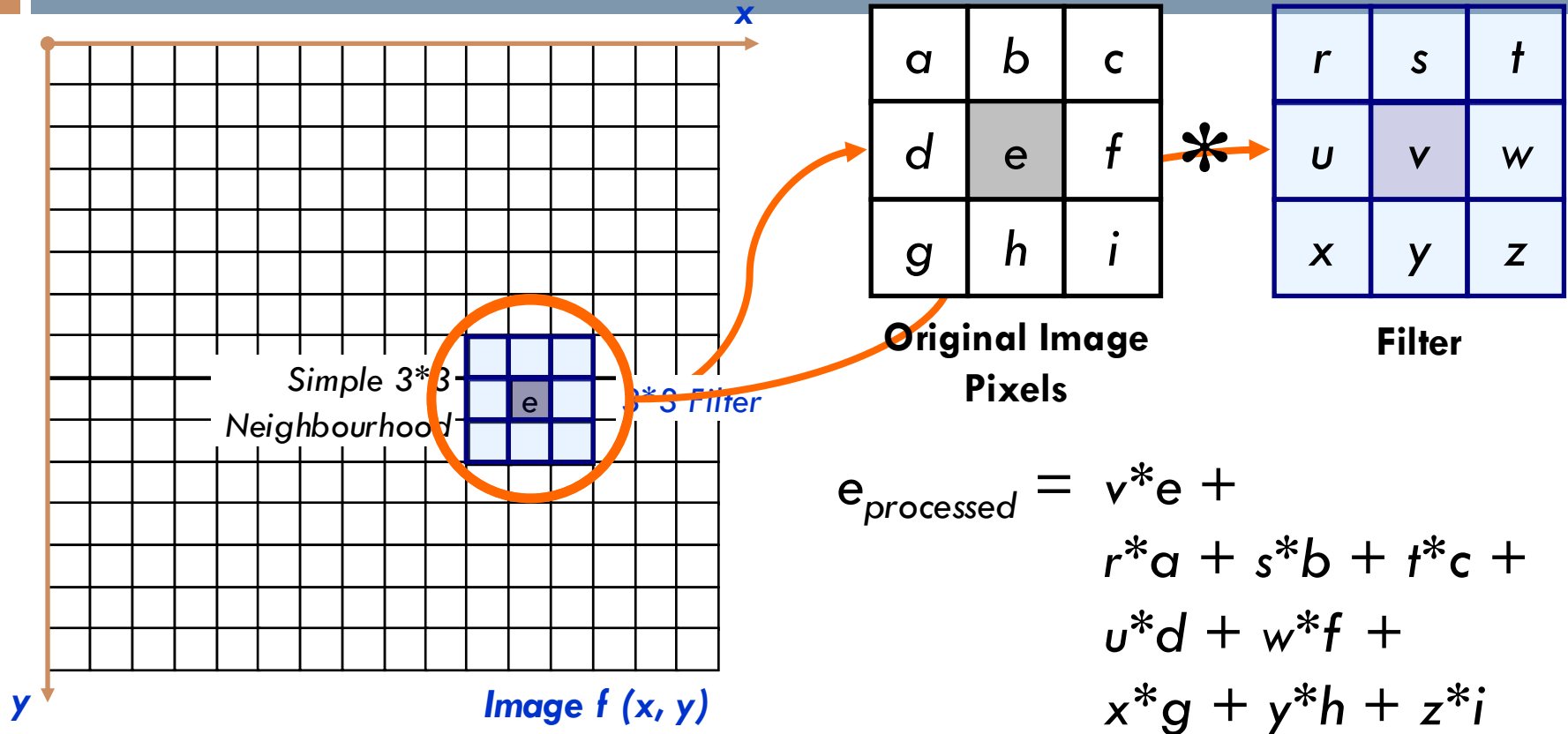
Mask coefficients

$$z_5 \leftarrow z = w_1 z_1 + w_2 z_2 + \dots + w_9 z_9 = \sum_{i=1}^9 w_i z_i$$

- z gives the output intensity value for the processed image (to be stored in a new array) at the location of z_5 in the input image

Local Operations through Spatial Filtering

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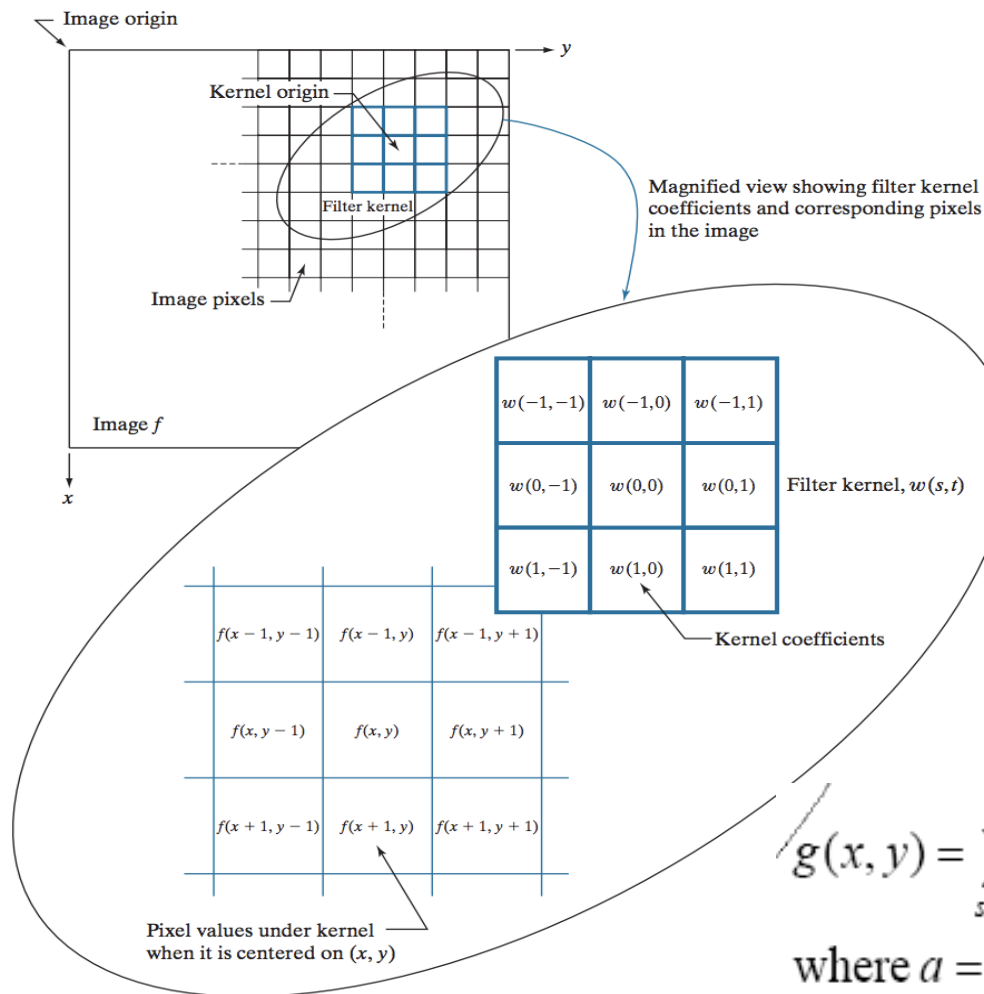


- The above is repeated for every pixel in the original image to generate the smoothed image

Local Operations through Spatial Filtering

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FIGURE 3.28
The mechanics of linear spatial filtering using a 3×3 kernel. The pixels are shown as squares to simplify the graphics. Note that the origin of the image is at the top left, but the origin of the kernel is at its center. Placing the origin at the center of spatially symmetric kernels simplifies writing expressions for linear filtering.



The mechanics of spatial filtering

For an image of size $M \times N$ and a mask of size $m \times n$

The resulting output gray level for any coordinates x and y is given by

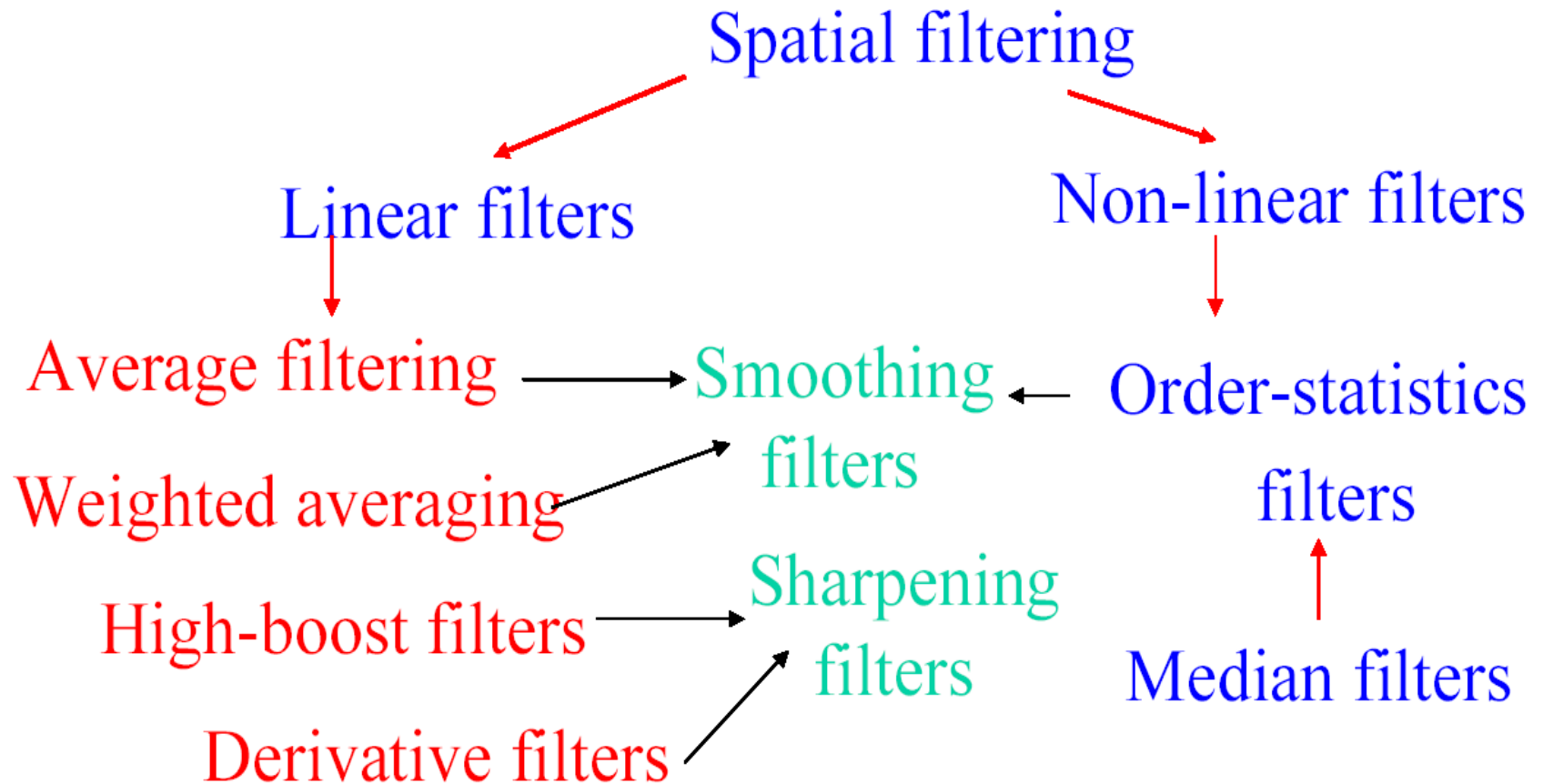
$$g(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x+s, y+t)$$

where $a = (m-1)/2$, $b = (n-1)/2$

$x = 0, 1, 2, \dots, M-1$, $y = 0, 1, 2, \dots, N-1$,

Types of Spatial Filtering

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Spatial Filtering for Smoothing

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- ❑ For blurring/noise reduction;
- ❑ Blurring is usually used in preprocessing steps, e.g., to remove small details from an image prior to object extraction, or to bridge small gaps in lines or curves
- ❑ **Equivalent to Low-pass spatial filtering in frequency domain** because smaller (high frequency) details are removed based on **neighborhood averaging** (averaging filters)
- ❑ **Implementation:**
 - ✓ The simplest form of the spatial filter for averaging is a square mask (assume $m \times m$ mask) with the same coefficients $1/m^2$ to preserve the gray levels (**averaging**).
- ❑ **Applications:** Reduce noise; smooth false contours
- ❑ **Side effect:** Edge blurring

Smoothing Spatial Filters

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- One of the simplest **spatial filtering** operations we can perform is a smoothing operation
- Simply average** all the pixels in a neighbourhood around a central value
- Especially useful in removing noise from images

a b

FIGURE 3.31

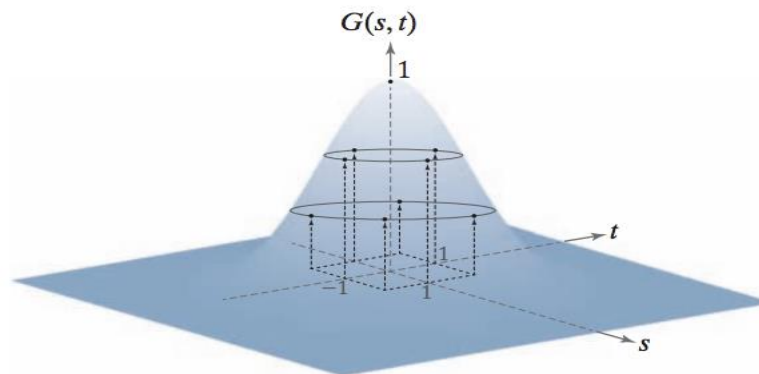
Examples of smoothing kernels: (a) is a *box* kernel; (b) is a *Gaussian* kernel.

$$\frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \frac{1}{4.8976} \times \begin{bmatrix} 0.3679 & 0.6065 & 0.3679 \\ 0.6065 & 1.0000 & 0.6065 \\ 0.3679 & 0.6065 & 0.3679 \end{bmatrix}$$

a b

FIGURE 3.35

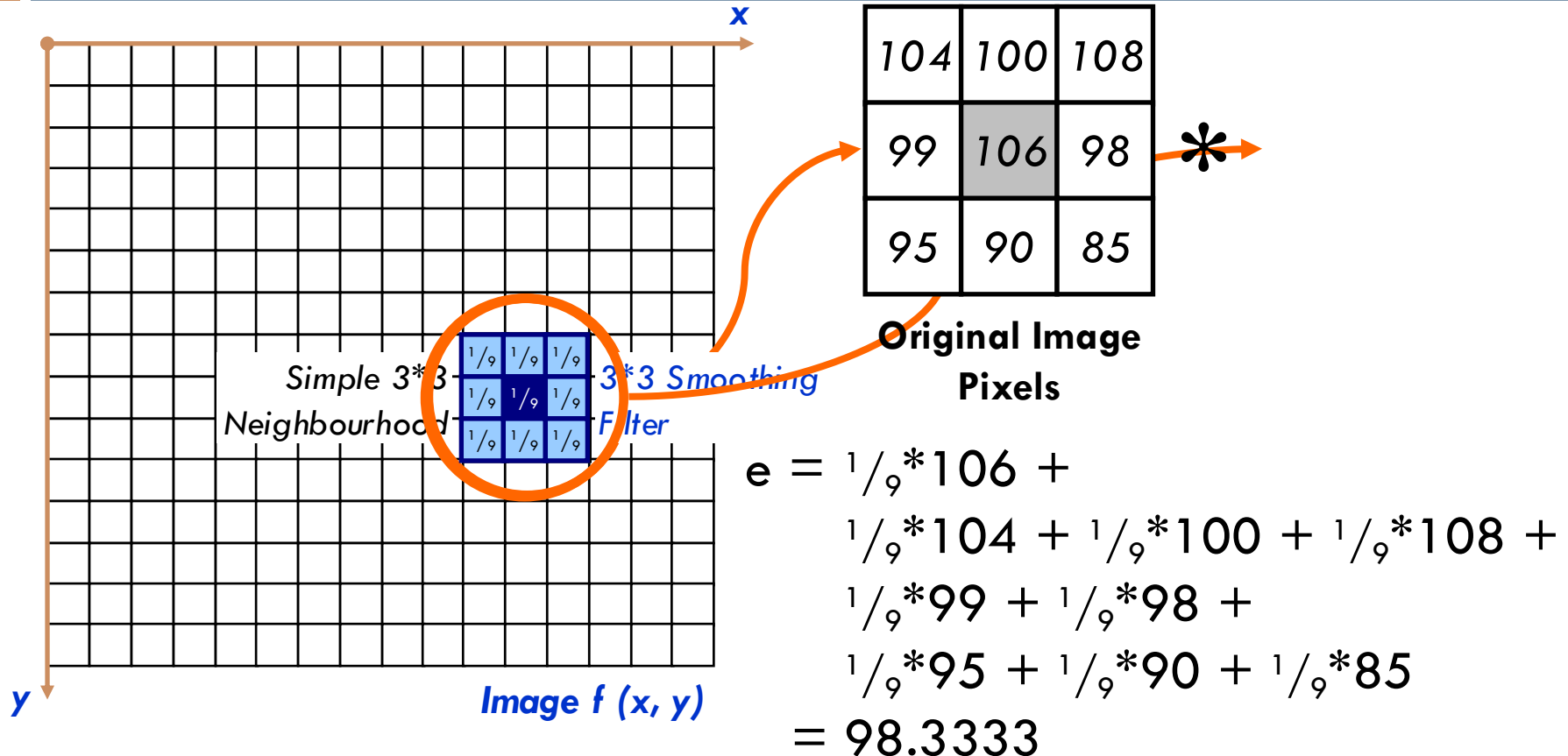
(a) Sampling a Gaussian function to obtain a discrete Gaussian kernel. The values shown are for $K = 1$ and $\sigma = 1$. (b) Resulting 3×3 kernel [this is the same as Fig. 3.31(b)].



$$\frac{1}{4.8976} \times \begin{bmatrix} 0.3679 & 0.6065 & 0.3679 \\ 0.6065 & 1.0000 & 0.6065 \\ 0.3679 & 0.6065 & 0.3679 \end{bmatrix}$$

Smoothing Spatial Filters

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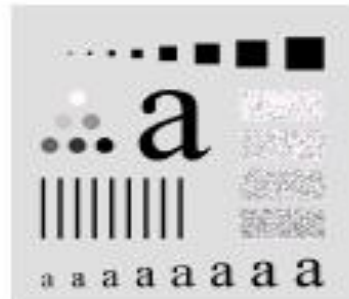


- The above is repeated for every pixel in the original image to generate the smoothed image

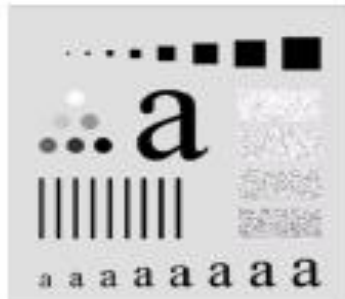
Spatial Filtering for Smoothing:: Example

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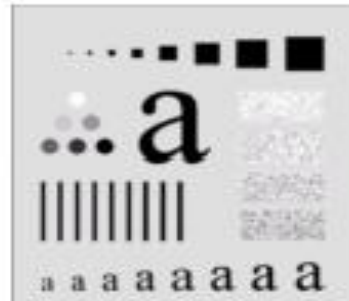
Original image
size: 500 x 500



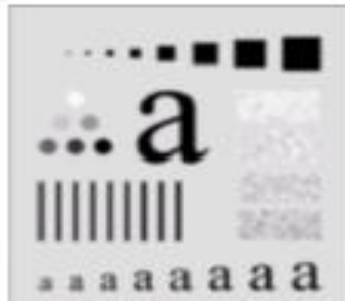
Smoothed by
3 x 3 box filter



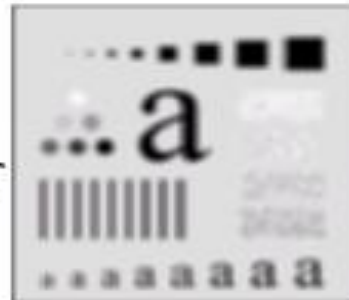
Smoothed by
5 x 5 box filter



Smoothed by
9 x 9 box filter



Smoothed by
15 x 15 box filter

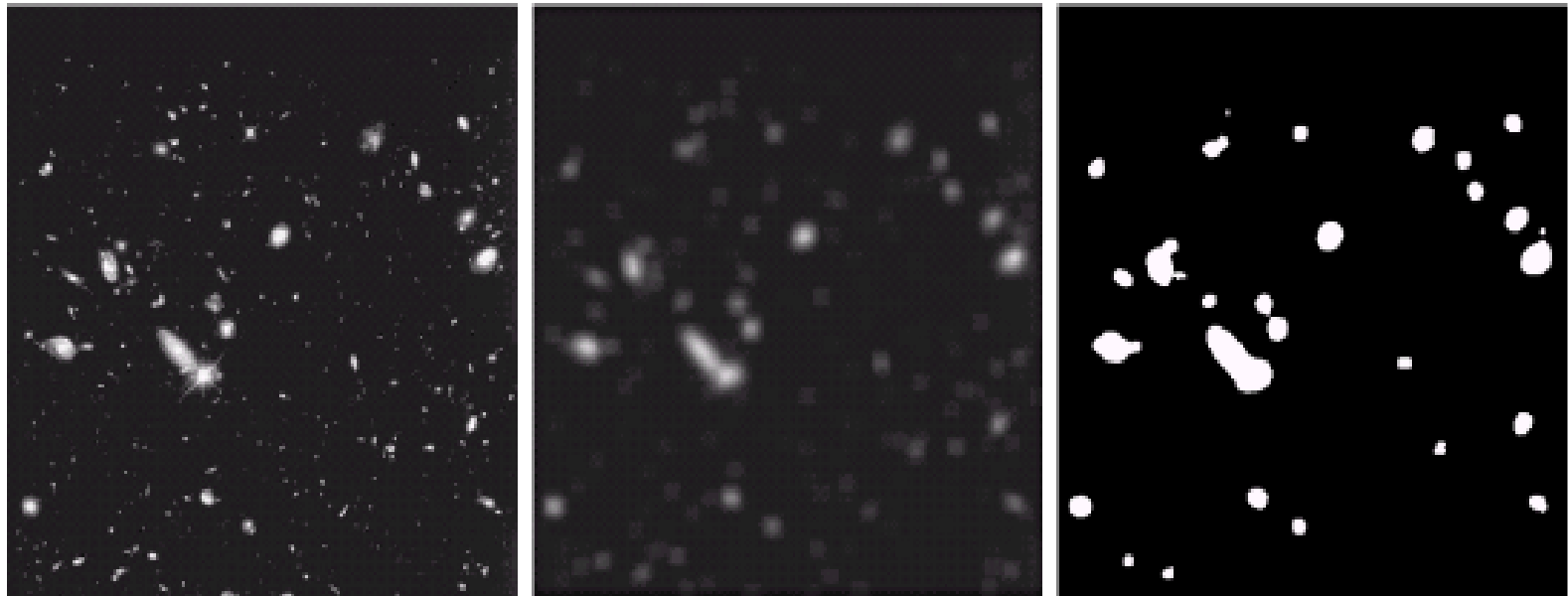


Smoothed by
35 x 35 box filter



Spatial Filtering for Smoothing:: Example

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a b c

FIGURE 3.36 (a) Image from the Hubble Space Telescope. (b) Image processed by a 15×15 averaging mask. (c) Result of thresholding (b). (Original image courtesy of NASA.)

Weighted Smoothing Spatial Filters

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- More effective smoothing filters can be generated by allowing different pixels in the neighbourhood different weights in the averaging function
 - ✓ Pixels **closer** to the central pixel are more important
 - ✓ Often referred to as **a weighted averaging**

$\frac{1}{16}$	$\frac{2}{16}$	$\frac{1}{16}$
$\frac{2}{16}$	$\frac{4}{16}$	$\frac{2}{16}$
$\frac{1}{16}$	$\frac{2}{16}$	$\frac{1}{16}$

Weighted Averaging Filter

Order Statistics Filtering

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- ❑ Nonlinear spatial filters
- ❑ Output is based on order of gray levels in the masked area (sub-image)
- ❑ Examples: Median filtering, Max & Min filtering
 - ✓ **Min:** Set the pixel value to the minimum in the neighbourhood
 - ✓ **Max:** Set the pixel value to the maximum in the neighbourhood
 - ✓ **Median:** The median value of a set of numbers is the midpoint value in that set (e.g. from the set [1, 7, 15, 18, 24] 15 is the median). Sometimes the median works better than the average

Order Statistics Filtering

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Median Filtering

- ❑ Assigns the mid value of all the gray levels in the mask to the center of mask;
- ❑ Particularly effective when
 - ✓ *the noise pattern consists of strong, spiky components (**salt-and-pepper**)*
 - ✓ *edges are to be preserved*
 - ✓ *Force points with distinct gray levels to be more like their neighbors*

10	20	20
20	15	20
20	25	100



Output = ? **20**

Order Statistics Filtering

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Minimum Filtering

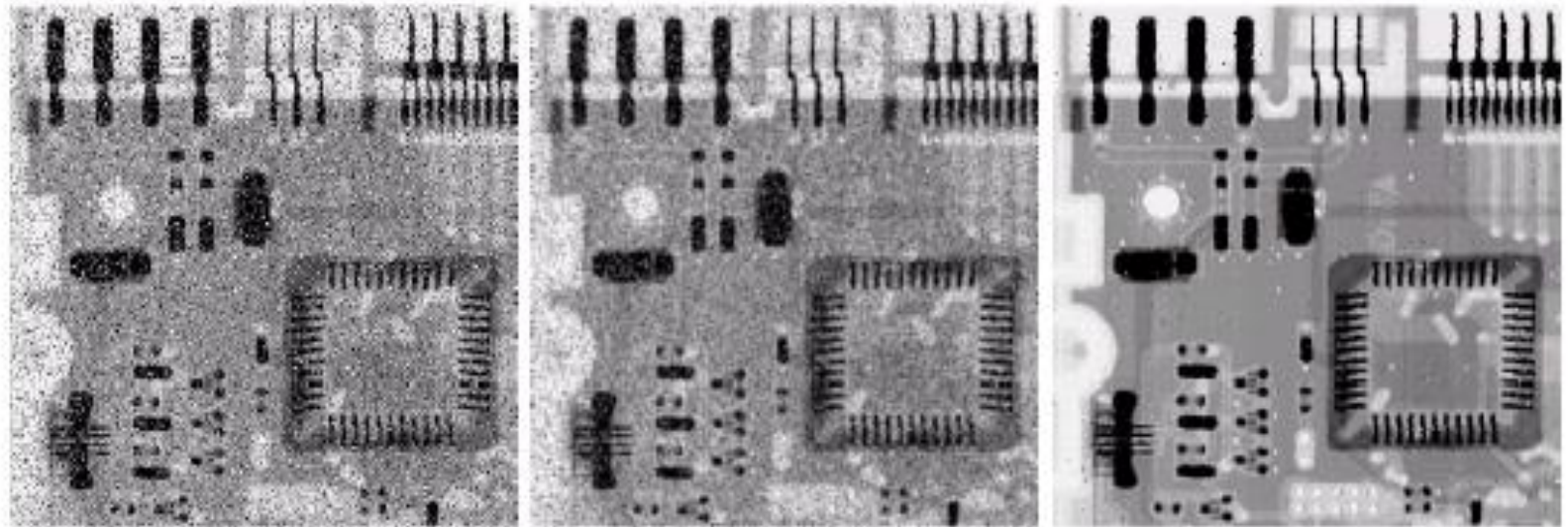
- ❑ Assigns the minimum value of all the gray levels in the mask to the center of mask;
- ❑ Particularly effective when
 - ✓ *Only Salt noise is present*

Maximum Filtering

- ❑ Assigns the maximum value of all the gray levels in the mask to the center of mask;
- ❑ Particularly effective when
 - ✓ *Only Pepper noise is present*
- ❑ *Edges are to be preserved*
- ❑ *Force points with distinct gray levels to be more like their neighbors*

Order Statistics Filtering

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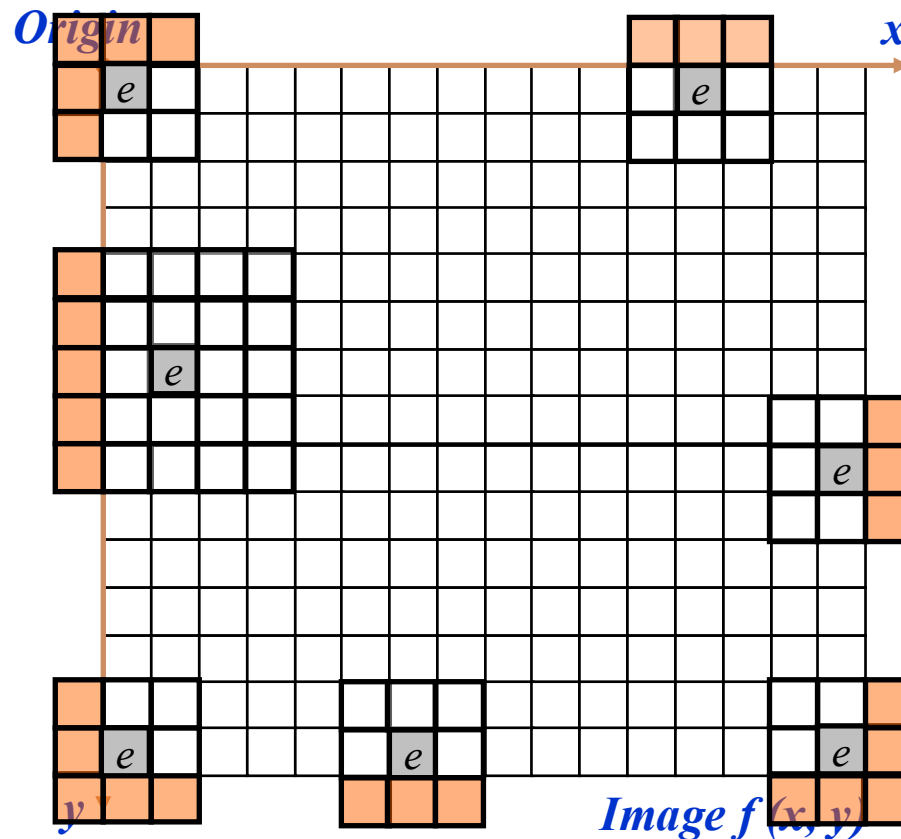
a b c

FIGURE 3.37 (a) X-ray image of circuit board corrupted by salt-and-pepper noise. (b) Noise reduction with a 3×3 averaging mask. (c) Noise reduction with a 3×3 median filter. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)

Strange Things Happen At The Edges!

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- At the edges of an image, we are missing pixels to form a neighbourhood



Border Pixels Treatment

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- ❑ Mask operation near the image border
- ❑ Problem arises when part of the mask is located outside the image plane; to handle the problem:
 - ✓ **Truncate the image**
 - Discard the problem pixels (e.g., 512x512 input 510x510 output, if mask size is 3x3)
 - ✓ **Image Padding**
 - expand the input image by padding zeros or 255(512x512 input 514x514output)
 - *Image padding is not good; creates artificial lines or edges on the border*

Border Pixels Treatment

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Replicate border pixels

- We normally use the gray levels of border pixels to fill up the expanded region (for 3x3 mask). For larger masks, a border region equal to half of the mask size is mirrored on the expanded region.

102	102	130	143	123	115	...	...
102	102	130	143	123	115	...	...
93	93	...	...	...			
98	98	...					
82	82	...					
65	65						
...	...						
...	...						

Expanded area

Original image size (shaded area)

Histogram

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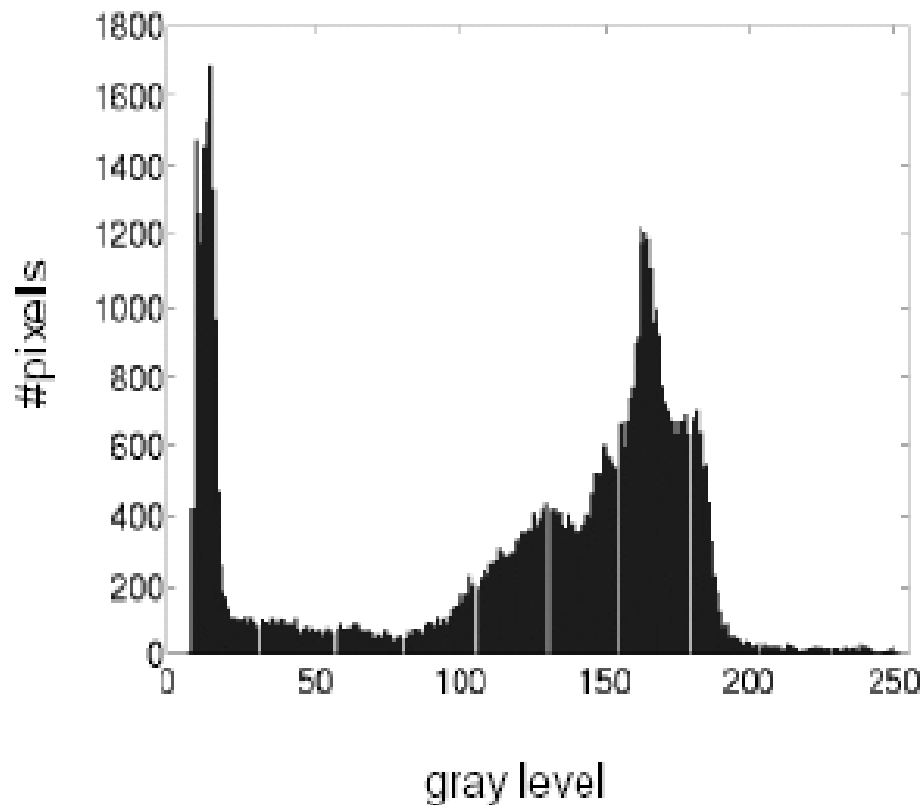
- The histogram of an image represents the frequency of pixels in an image
- The histogram of an image shows the distribution of gray levels in the image
- Massively useful in image processing especially in image segmentation
- Histogram of an image with gray level (0 to L-1):

A discrete function $h(r_k) = n_k$, where r_k is the k^{th} gray level and n_k is the number of pixels in the image having gray level r_k .

- How to obtain histogram?
 - ✓ For B bit image, initialize 2^B counters with 0
 - ✓ Loop over all pixels x,y
 - ✓ When encountering gray level $f(x,y)=l$, increment counter # l

Histogram : Example

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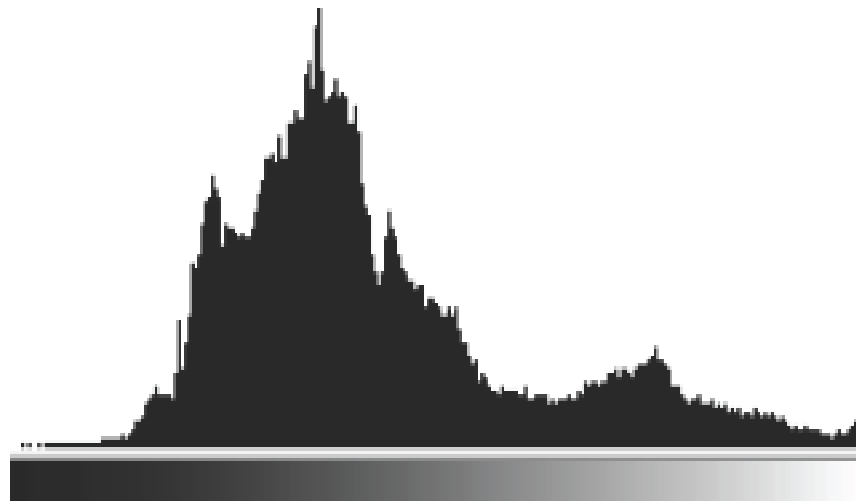


*Cameraman
image*

Histogram and Image Brightness/Contrast

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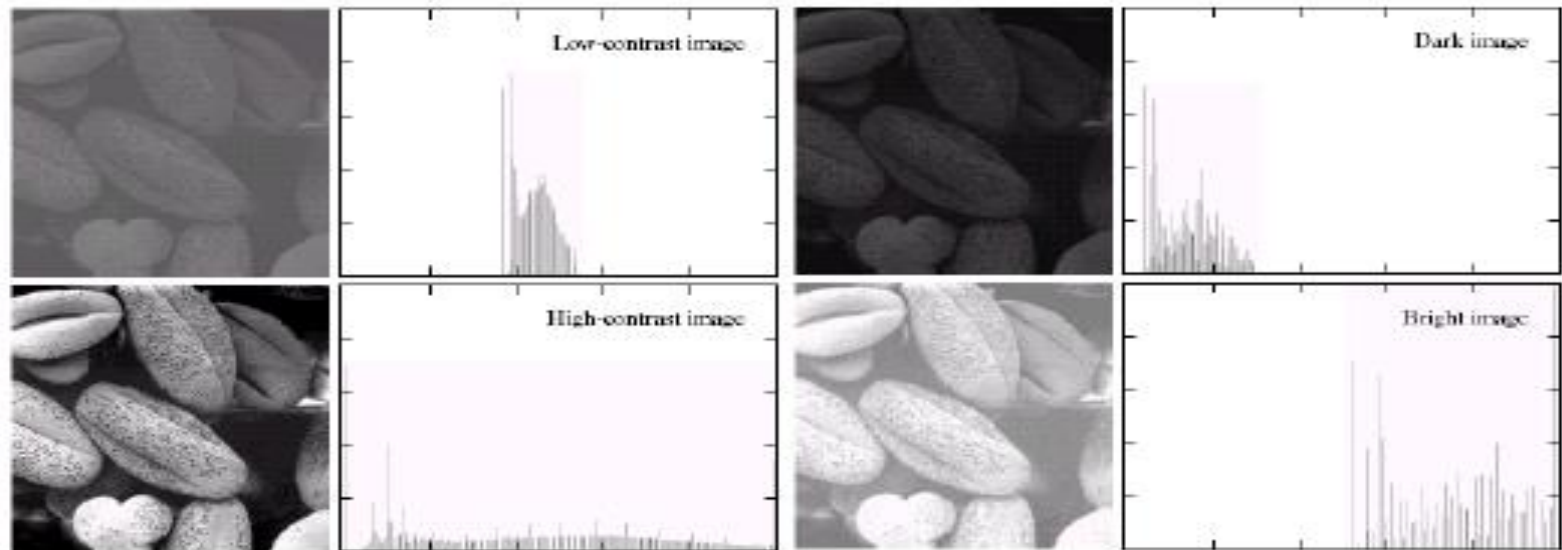
- ❑ Changing the size of the image shrinks or stretches the histogram vertically.
- ❑ Adjusting brightness shifts the histogram horizontally.
- ❑ Adjusting contrast causes the histogram to stretch or shrink horizontally:
 - ✓ **Stretching = more contrast**
 - ✓ **Shrinking = less contrast**



Histogram : Example

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- ❑ A selection of images and their Histograms
- ❑ Note that the high contrast image has the most evenly spaced histogram
- ❑ Histograms of low contrast images are located in certain portions and not in the entire gray scale range

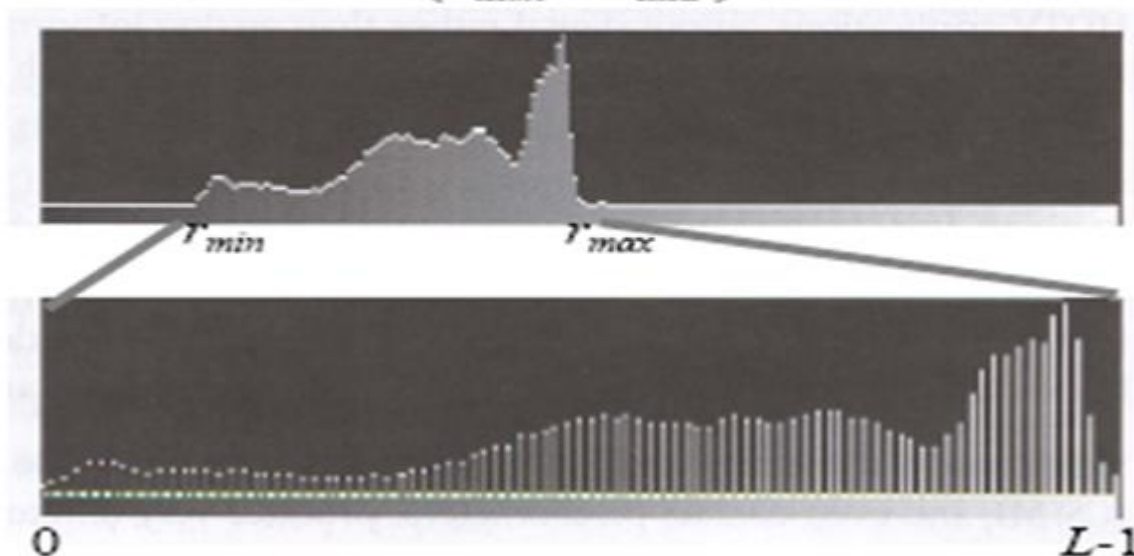


Contrast Stretching through Histogram

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- If r_{max} and r_{min} are the maximum and minimum gray level of the input image and L is the total gray levels of output image The transformation function for contrast stretching will be

$$s = T(r) = (r - r_{min}) \left(\frac{L - 1}{r_{max} - r_{min}} \right)$$



Histogram Equalization

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- The idea is to find a transformation $s=T(r)$ to be applied to each pixel of the input image $f(x,y)$ such that a uniform distribution of gray levels in the entire range results for the output image $g(x,y)$

The discrete approximation of the transformation function for histogram equalization is:

$$s_k = T(r_k) = \sum_{j=0}^k p_r(r_j) \quad \text{for } 0 \leq k \leq L-1$$

where $p_r(r_j) = \frac{n_j}{n}$, $j = 0, \dots, L-1$ and $n = \sum_{j=0}^{L-1} n_j$

n_j : number of pixels with gray level r_j

n : total number of pixels

Histogram Equalization

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- ❑ Spreading out the frequencies in an image (or equalising the image) is a simple way to improve dark or washed-out images
- ❑ The formula for histogram equalization is given where,

- ✓ r_k : input intensity
- ✓ s_k : processed intensity
- ✓ k : the intensity (e.g 0.0 – 1.0)
- ✓ n_j : the frequency of intensity j
- ✓ n : the sum of all frequencies

$$\begin{aligned} s_k &= T(r_k) \\ &= \sum_{j=1}^k p_r(r_j) \\ &= \sum_{j=1}^k \frac{n_j}{n} \end{aligned}$$

Histogram Equalization

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52	55	61	66	70	61	64	73
63	59	55	90	109	85	69	72
62	59	68	113	144	104	66	73
63	58	71	122	154	106	70	69
67	61	68	104	126	88	68	70
79	65	60	70	77	68	58	75
85	71	64	59	55	61	65	83
87	79	69	68	65	76	78	94

Value	Count	Value	Count	Value	Count	Value	Count	Value	Count
52	1	64	2	72	1	85	2	113	1
55	3	65	3	73	2	87	1	122	1
58	2	66	2	75	1	88	1	126	1
59	3	67	1	76	1	90	1	144	1
60	1	68	5	77	1	94	1	154	1
61	4	69	3	78	1	104	2		
62	1	70	4	79	2	106	1		
63	2	71	2	83	1	109	1		

Histogram Equalization

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52	55	61	66	70	61	64	73
63	59	55	90	109	85	69	72
62	59	68	113	144	104	66	73
63	58	71	122	154	106	70	69
67	61	68	104	126	88	68	70
79	65	60	70	77	68	58	75
85	71	64	59	55	61	65	83
87	79	69	68	65	76	78	94

Initial Image

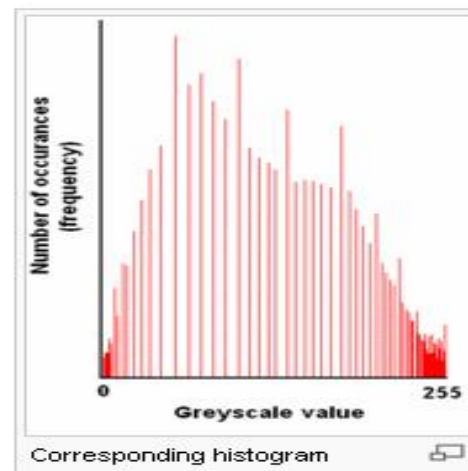
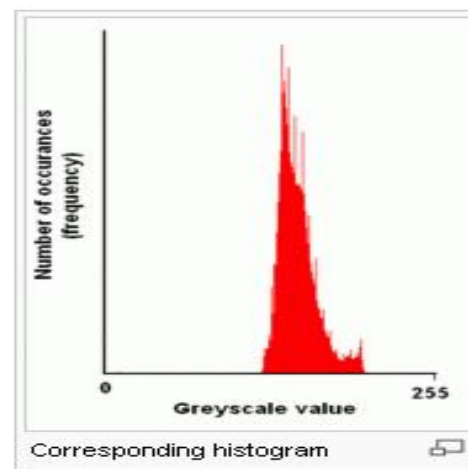
0	12	53	93	146	53	73	166
65	32	12	215	235	202	130	158
57	32	117	239	251	227	93	166
65	20	154	243	255	231	146	130
97	53	117	227	247	210	117	146
190	85	36	146	178	117	20	170
202	154	73	32	12	53	85	194
206	190	130	117	85	174	182	219

Image After Equalization

Notice that the minimum value (52) is now 0 and the maximum value (154) is now 255.

Histogram Equalization

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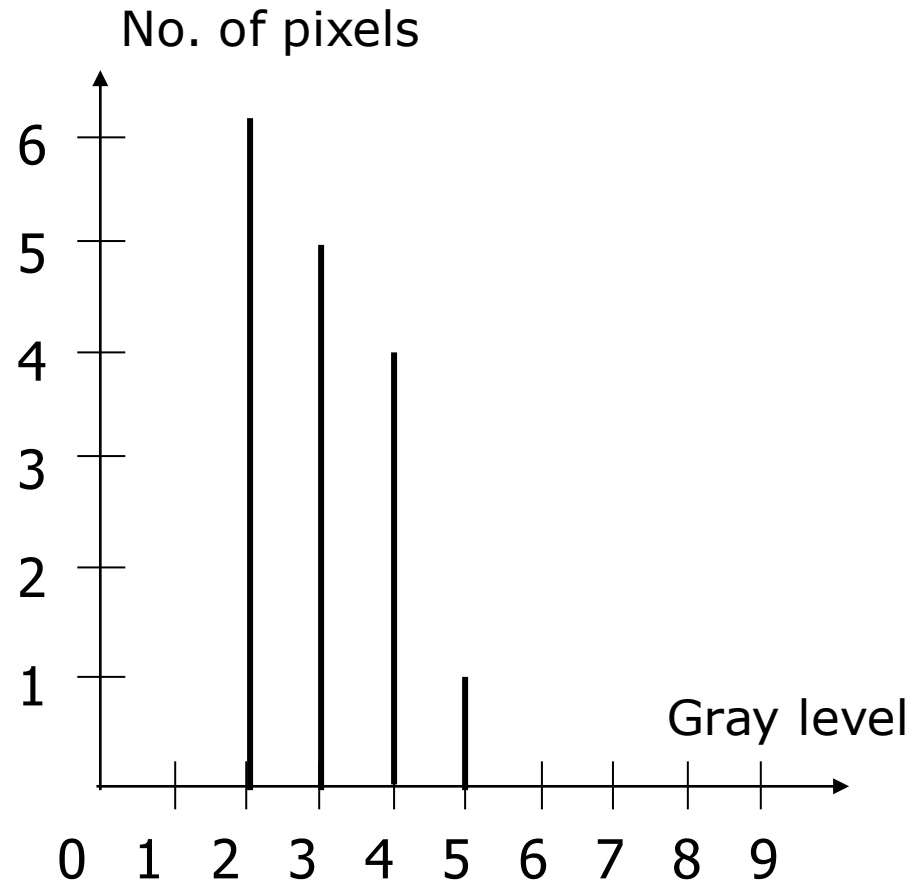
Histogram Equalization

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2	3	3	2
4	2	4	3
3	2	3	5
2	4	2	4

4x4 image

Gray scale = [0,9]



histogram

Histogram Equalization

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Gray Level(j)	0	1	2	3	4	5	6	7	8	9
No. of pixels	0	0	6	5	4	1	0	0	0	0
PDF	0	0	$\frac{6}{16}$	$\frac{5}{16}$	$\frac{4}{16}$	$\frac{1}{16}$	$\frac{0}{16}$	$\frac{0}{16}$	$\frac{0}{16}$	$\frac{0}{16}$
CDF	0	0	$\frac{6}{16}$	$\frac{11}{16}$	$\frac{15}{16}$	$\frac{16}{16}$	1	1	1	1
$s \times 9$	0	0	$\frac{3.3}{\approx 3}$	$\frac{6.1}{\approx 6}$	$\frac{8.4}{\approx 8}$	9	9	9	9	9

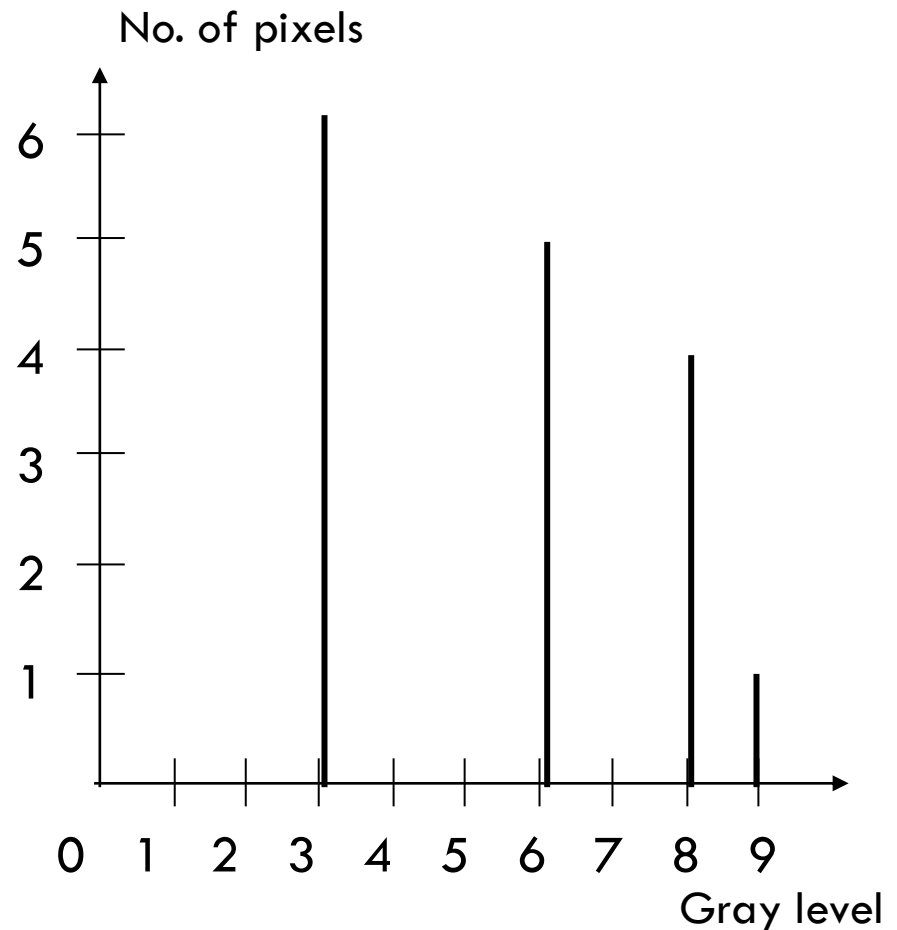
Histogram Equalization

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3	6	6	3
8	3	8	6
6	3	6	9
3	8	3	8

Output image

Gray scale = [0,9]



Equalized histogram

Example: Histogram Equalization

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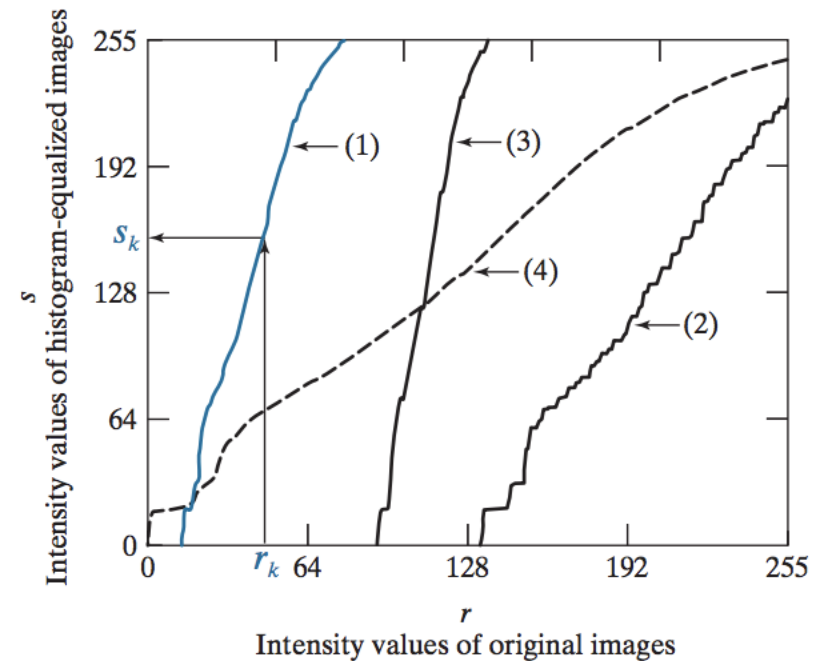
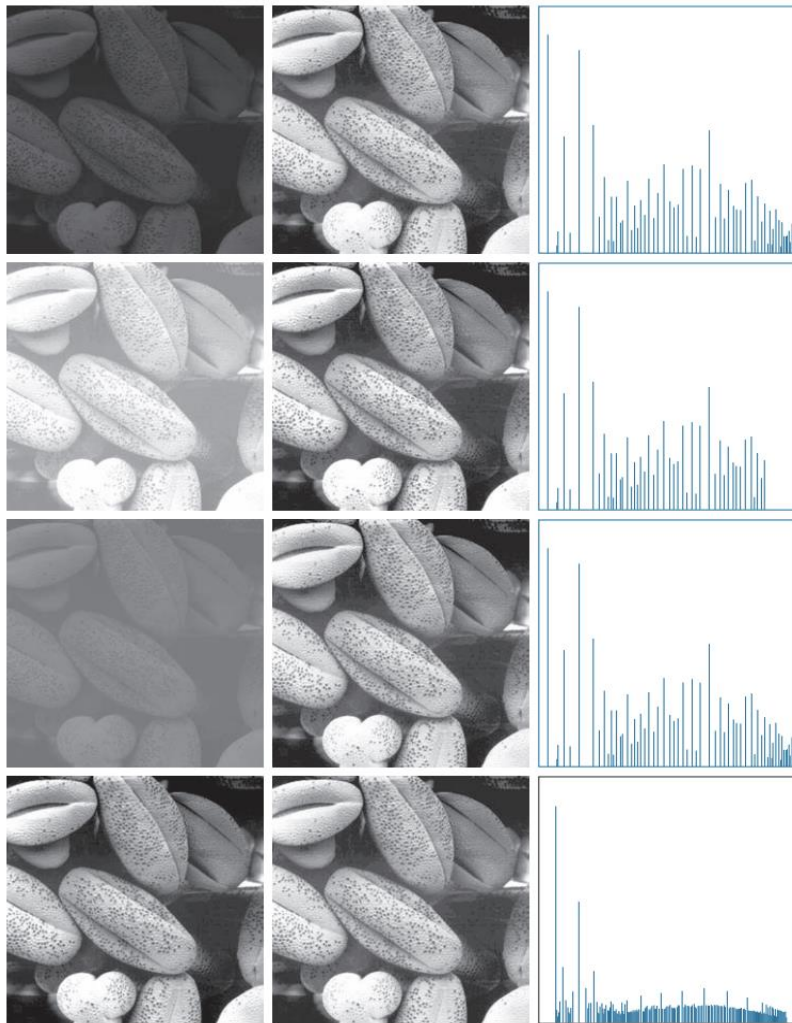


FIGURE 3.20 Left column: Images from Fig. 3.16. Center column: Corresponding histogram-equalized images. Right column: histograms of the images in the center column (compare with the histograms in Fig. 3.16).

Local Enhancement through Histogram Processing

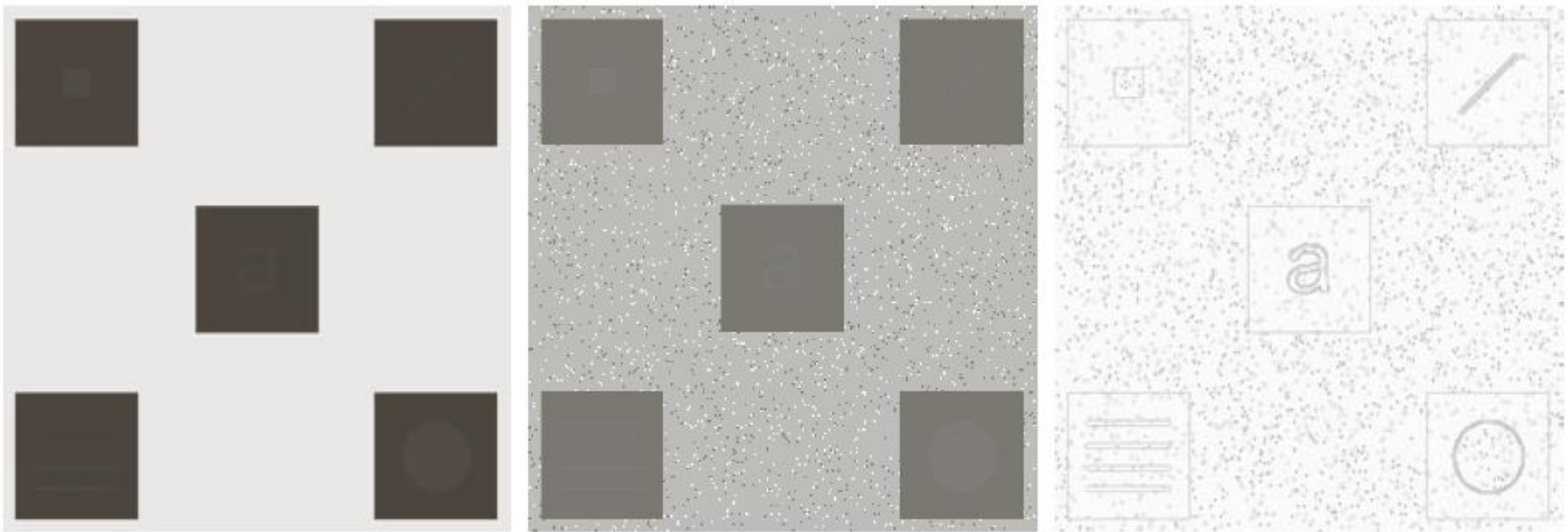
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- ❑ The histogram processing methods discussed earlier are global in the sense that pixels are modified by a transformation function based on the gray level content of the entire image
- ❑ Although this global approach is suitable for overall enhancement. There are cases in which it is necessary to enhance details over small areas in the image
- ❑ The histogram equalization can be easily adapted to local enhancement
- ❑ The procedure is to define a neighborhood and move the center of this area from pixel to pixel
- ❑ At each location, the histogram of the points in the neighborhood is computed and then histogram transformation function is obtained

Local Enhancement through Histogram Processing

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- Map the intensity of the pixel centered in the neighborhood
- Move to the next location and repeat the procedure



a b c

FIGURE 3.26 (a) Original image. (b) Result of global histogram equalization. (c) Result of local histogram equalization applied to (a), using a neighborhood of size 3×3 .

Use of Histogram Statistics for Image Enhancement

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Average Intensity

$$m = \sum_{i=0}^{L-1} r_i p(r_i) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y)$$

Variance

$$\sigma^2 = \sum_{i=0}^{L-1} (r_i - m)^2 p(r_i) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [f(x, y) - m]^2$$

Use of Histogram Statistics for Image Enhancement

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Local average intensity

$$m_{s_{xy}} = \sum_{i=0}^{L-1} r_i p_{s_{xy}}(r_i)$$

s_{xy} denotes a neighborhood

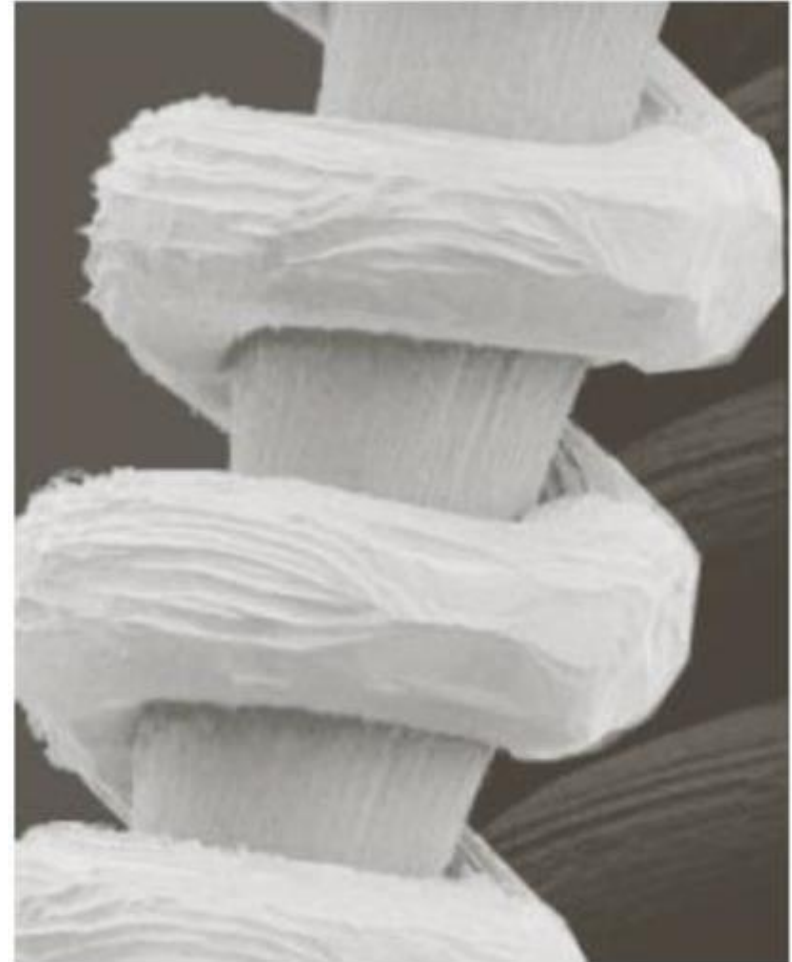
Local variance

$$\sigma_{s_{xy}}^2 = \sum_{i=0}^{L-1} (r_i - m_{s_{xy}})^2 p_{s_{xy}}(r_i)$$

Use of Histogram Statistics for Image Enhancement

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- ❑ The statistical parameters can be used in various ways
- ❑ Enhance the background filament
- ❑ Enhance details in dark areas while leaving light area unchanged.
- ❑ Define rules to chose the candidate pixels that need to be enhanced



Use of Histogram Statistics for Image Enhancement

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- ❑ A pixel at point(x,y) is considered if:
 - ✓ $m_{sxy} \leq k_0 \times M_G$, where k_0 is a positive constant less than 1.0, and M_G is global mean
 - ✓ $\sigma_{sxy} \leq k_2 \times D_G$, where D_G is the global standard deviation and k_2 is a positive constant
 - ✓ Also need to put a lower limit on SD to avoid distorting areas which don't have details, i.e., $k_1 D_G \leq \sigma_{sxy}$, with $k_1 < k_2$

Use of Histogram Statistics for Image Enhancement

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- ❑ A pixel that meets all above conditions is processed simply by multiplying it by a specified constant, E , to increase or decrease the value of its gray level relative to the rest of the image.
- ❑ The values of pixels that don't meet the enhancement conditions are left unchanged.

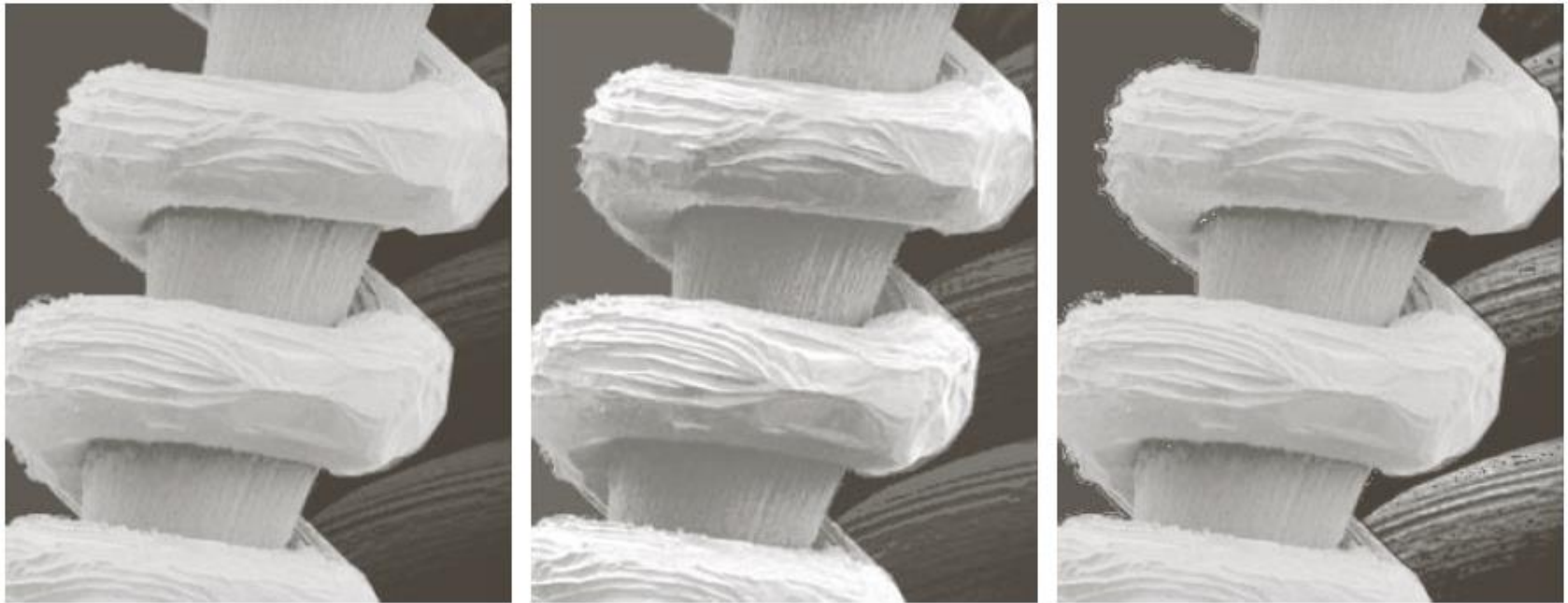
$$g(x, y) = \begin{cases} E \cdot f(x, y), & \text{if } m_{s_{xy}} \leq k_0 m_G \text{ and } k_1 \sigma_G \leq \sigma_{s_{xy}} \leq k_2 \sigma_G \\ f(x, y), & \text{otherwise} \end{cases}$$

m_G : global mean; σ_G : global standard deviation

$k_0 = 0.4$; $k_1 = 0.02$; $k_2 = 0.4$; $E = 4$

Use of Histogram Statistics for Image Enhancement

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a b c

FIGURE 3.27 (a) SEM image of a tungsten filament magnified approximately 130 $\times$. (b) Result of global histogram equalization. (c) Image enhanced using local histogram statistics. (Original image courtesy of Mr. Michael Shaffer, Department of Geological Sciences, University of Oregon, Eugene.)

References

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1. DIP by Gonzalez

For any query Feel Free to ask



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